

The Invisible Ledger

Quantifying the Invisible Wedge in Indonesia's Platform Economy

隱形帳簿：量化印尼平台經濟中的隱形缺口

Progress update: what has changed since the proposal oral of 1 October

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Since 1 October: what stayed, was added, changed

“The question and hypotheses are the ones you approved. The answers are new.”

STAYED

- The title, the research question, H1 and H2, word for word
- The measures: the invisible wedge, the ecosystem ratio, growth divergence
- The Tanah Abang opening and the three users of the numbers

ADDED

- BPS's own survey answers, three rounds (bought through SILASTIK)
- 27 foreign platforms in the comparison, up from 8
- A market-value test, and tests written down before the data were opened
- Primary texts of the 2023-26 rules

CHANGED

- The thesis now answers the oral deck's own question
- The wedge has a job: it helps explain why revenue swings
- H2's meaning: about half the 2023 rise in sellers was not first-time entry
- The market test came back only partly

Nothing the committee approved has moved. Everything promised at the oral has a result.

The oral deck's question, now answered

“How much commerce had actually moved online? It depends on which number you read.” (oral deck, slide 3)

THE ANSWER, IN SIX SENTENCES

In 2023 Indonesia argued about whether online selling was growing so fast it was hurting market traders, and the government shut TikTok Shop. The official figures for that year later said online sales rose 41%. I looked inside the survey behind those figures. Most sellers said their sales were flat or down, and their main complaint was a lack of customers. About half the new sellers weren't first-time sellers: the extra count leaned toward older, long-running small businesses selling through WhatsApp and social media. On the big platforms, revenue grew mostly because they charged more, not because people bought more.

Short version: less commerce had moved online in 2023 than the headlines suggested. Platform income followed prices; the official seller count rose by more than first-time entry can explain.

How the thesis now reads

“Start where everyone is, prove it where the numbers break, end with what anyone can check.”



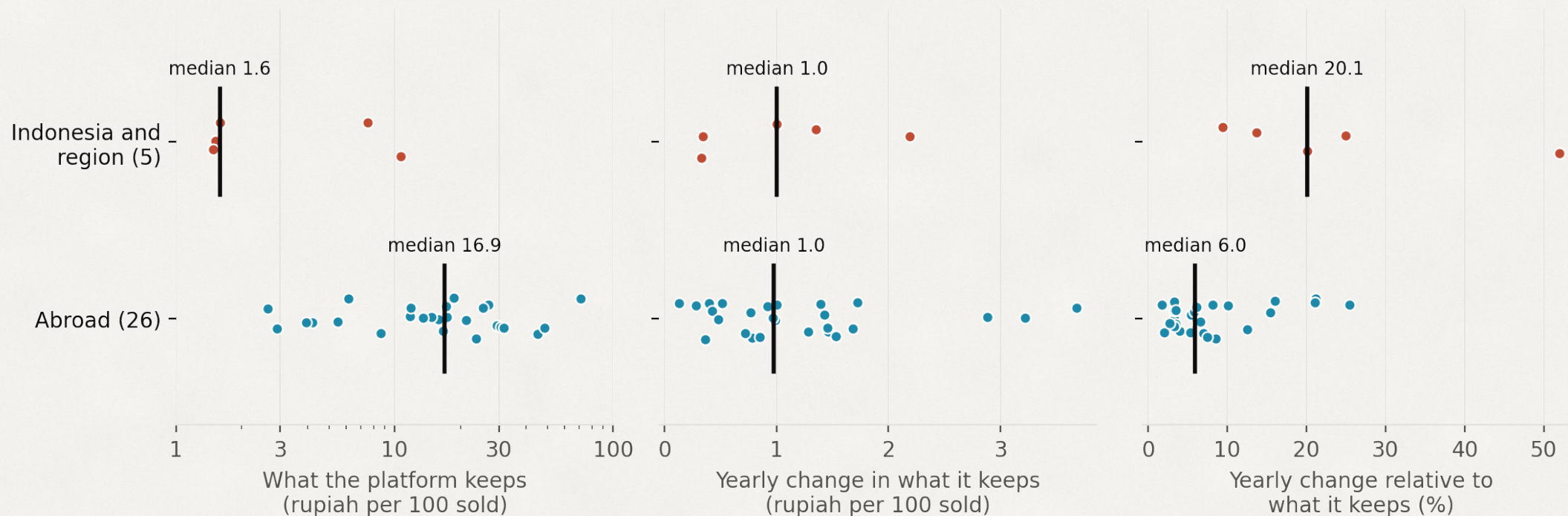
What the oral deck promised, and what came back

“Every promise has a result, including the ones that did not go our way.”

Promised at the oral	Result	Status
Thicken H1 with quarterly pairs	Does not hold: the quarterly data mostly cover the calmer years after the repricing	Failed, reported
A longer foreign comparison	27 foreign platforms; Indonesia still stands apart ($p = 0.004$; worst leave-one-out 0.016)	Done
Diagnose each large divergence	Grab mostly repricing; Blibli partly travel mix; GoTo cut discounts in 2024; Tokopedia 60.6% from fewer discounts	Done
Reconcile platform and BPS figures	They agree on growth where they cover the same sellers; levels cannot be fully reconciled, and the reasons are named	Done, with limits
Market test: does the market tell the two kinds of revenue growth apart?	Market value moved with buying (28 firms, $p = 0.002$); a different price for fee-driven revenue is not shown ($p = 0.20$)	Half answered
H2 caveat: “an arithmetic decomposition, not entry dynamics”	Entry test written down before the data: entry was too small; about half the 2023 rise was not first-time entry	Done, new centre

The wedge's job: similar price changes, thinner slice

"In rupiah, Indonesian platforms' changes were close to those abroad. Their slice is ten times thinner."

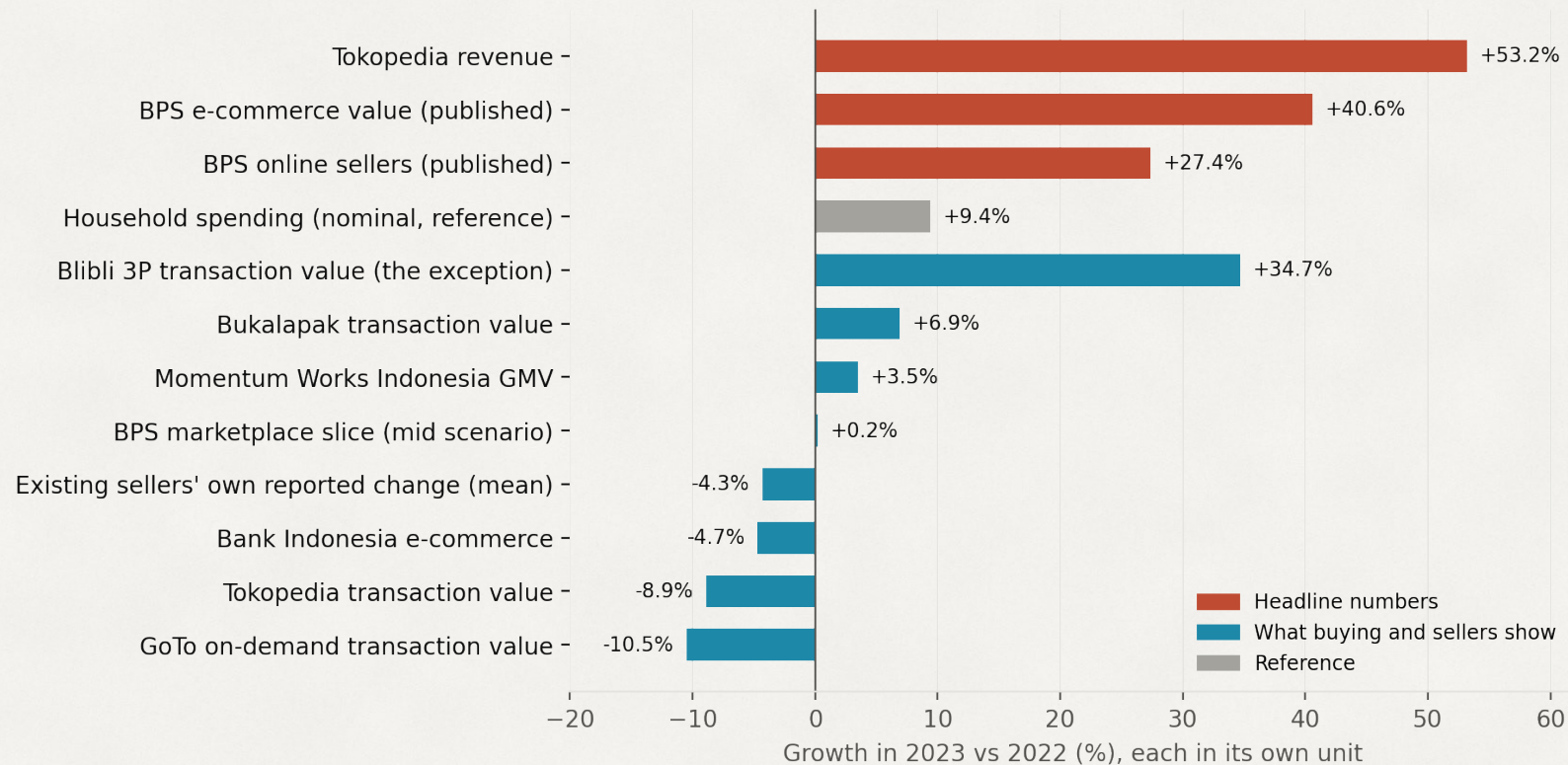


Reading it: a typical platform abroad keeps about 17 rupiah of every 100 sold; Indonesian marketplaces keep under 2. Both change what they keep by about 1 rupiah per 100 a year. On a 2-rupiah slice that is a huge change, so revenue growth swings about three times as much. H1 is still rejected; a thin slice helps explain why.

Issuer filings, 5 Indonesian and regional vs 26 foreign platforms (firm medians); in points $p = 0.65$, relative $p = 0.003$. Exploratory, 9 Oct.

2023: the headline numbers boomed, buying did not

“Where two numbers look at the same place, they agree. They split where only one of them looks.”



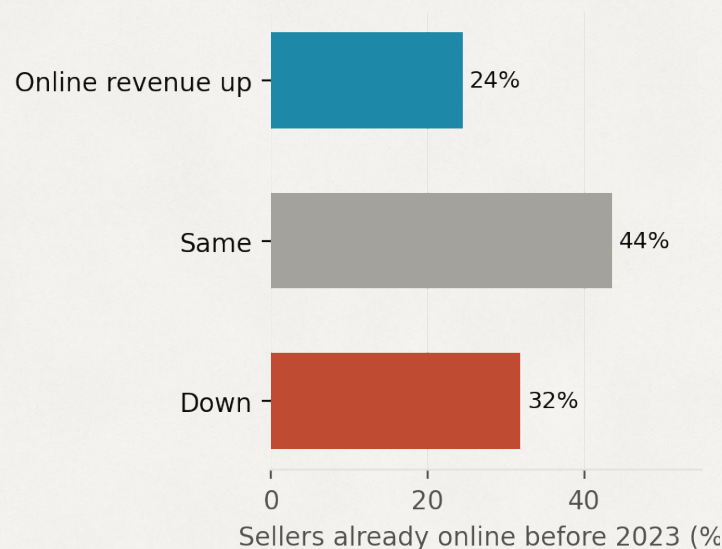
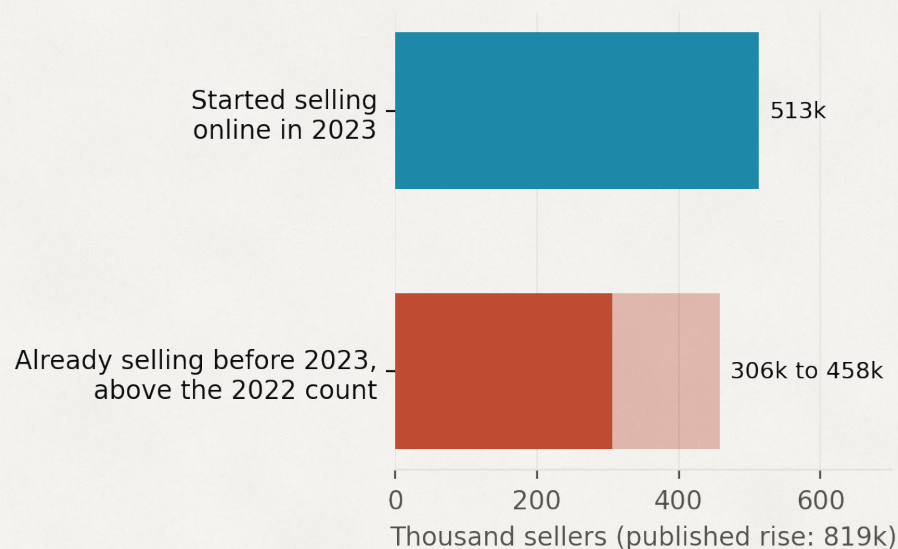
WHAT IT SHOWS

- Five of six Indonesia-only measures of buying grew less than household spending
- Existing sellers: 24% up, 44% same, 32% down
- Main complaint: not enough customers (41%, up from 35%)
- Bank Indonesia -5%, BPS +41%: they agree on the marketplaces and split outside them

Company filings; Bank Indonesia; Momentum Works; BPS publications and survey microdata. Growth 2023 vs 2022, each in its own unit.

H2 holds on paper; half the rise was not new entry

"If 820,000 people had started selling in 2023, they would say so. About 513,000 did."



THE ENTRY TEST

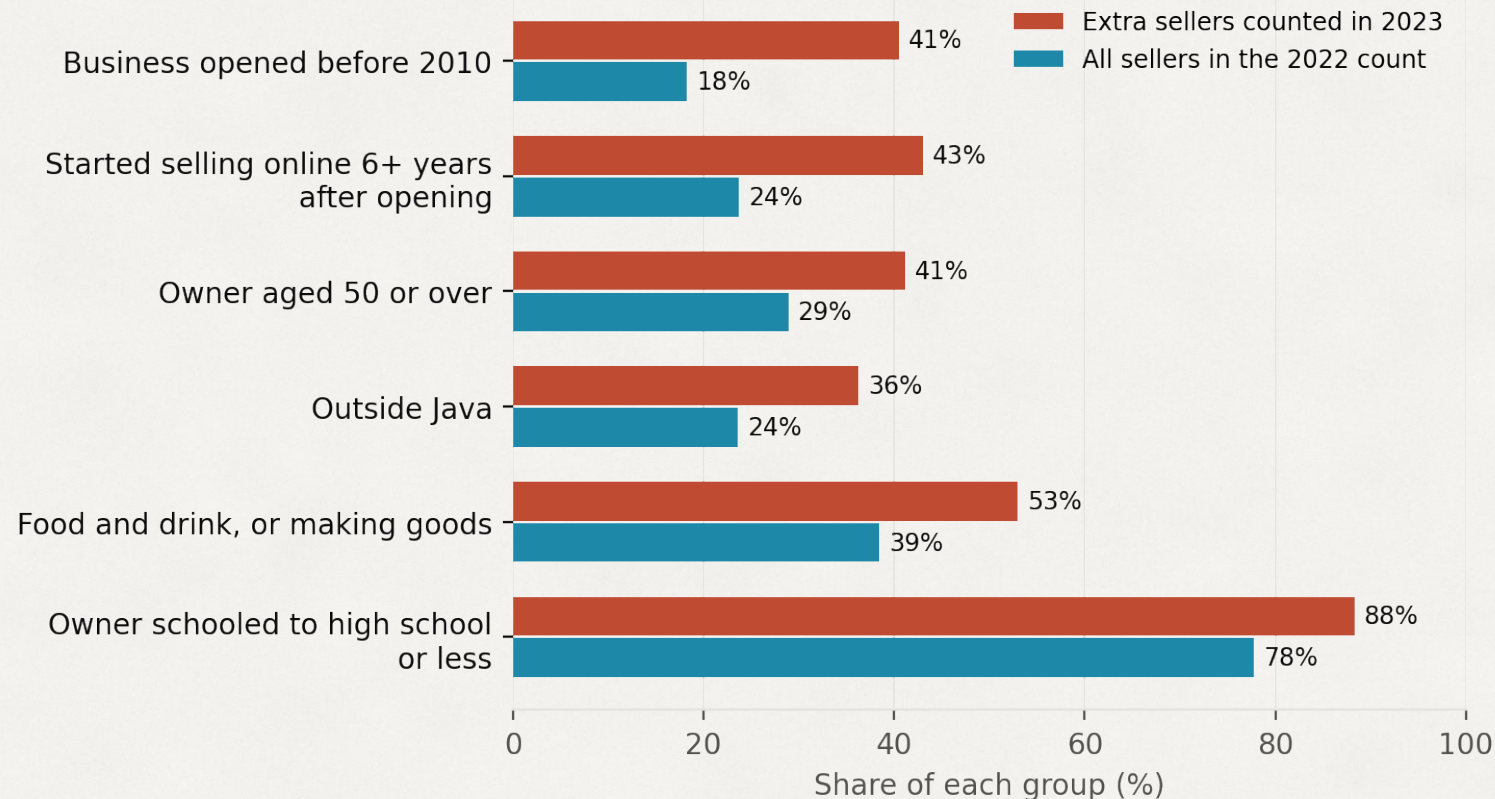
- Written down before the data: at least 1 in 5 sellers should say they started in 2023
- Result: about 1 in 7
- 37-56% of the rise was sellers already selling before 2023
- Large in the sampling-error scenarios checked; follows where the survey grew; 2024 is normal again

H2 on BPS's published numbers: holds (71% of 2022-23 growth from more businesses). The survey's own answers change what "more businesses" means.

BPS e-commerce survey microdata (2022 and 2023 rounds), weighted. Range depends on how many sellers stopped in 2023.

Who the extra sellers were

“The larger count was concentrated among older, long-running businesses selling through chat and social media.”



WHAT IT MEANS

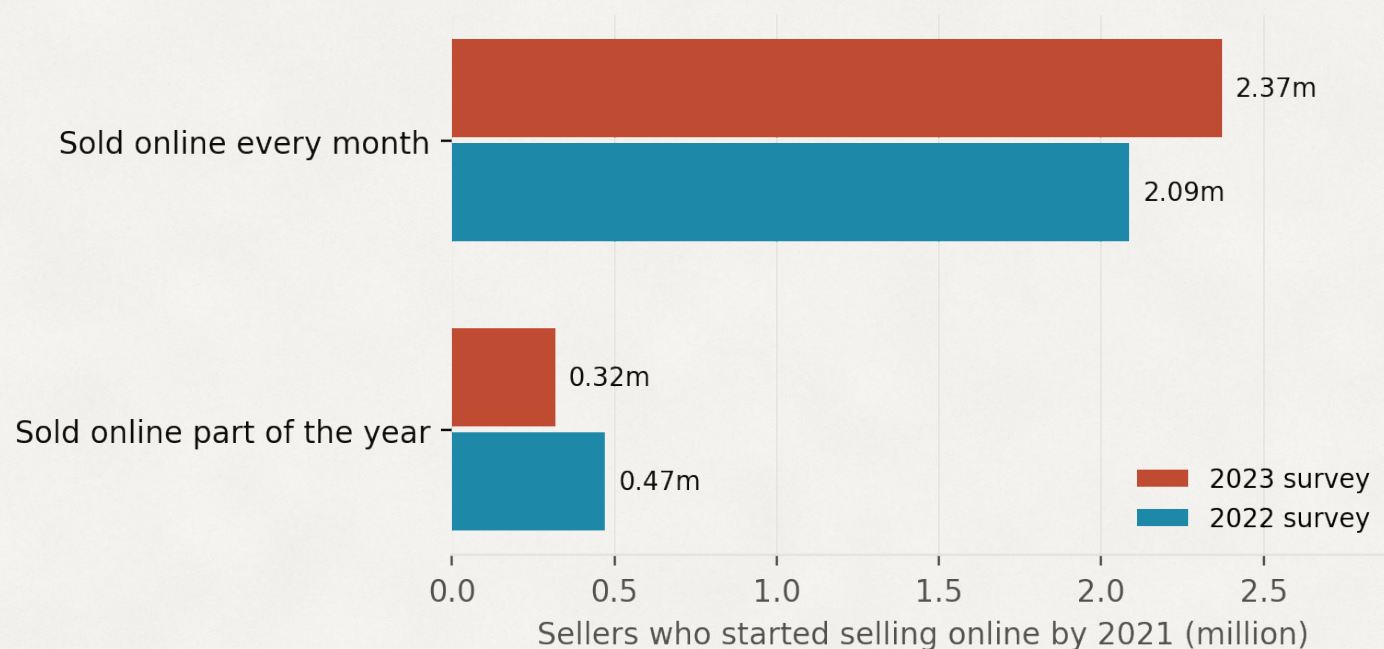
- Businesses opened before 2010, often going online years later
- Older owners, mostly schooled to high school or less
- More often outside Java; food stalls and small workshops
- Selling only through chat and social media

These are the kind of businesses the original question was about: real, long-running, and easy for the numbers to miss.

BPS e-commerce survey microdata (2020, 2022, 2023 rounds), weighted; about 450,000 sellers beyond normal change. Exploratory, 9 Oct.

Sellers back from a break: checked, not closed

“Part-year selling fell, even within the same start-year groups.”



THREE CHECKS

Months: among sellers who started by 2021, part-year selling fell from 472k to 318k; it falls for every start-year group.

Timing: returning happens every year; it would have to spike in 2023 alone, while 2020-22 groups shrank and 2024 is normal.

Who restarts: small firms close most often when young, with young owners (McKenzie and Paffhausen 2019). The extra count leans the other way. Returns stay the open alternative.

BPS survey microdata; the month question changed format between rounds. If recall errors run forward, as in Neter and Waksberg (1964), the entry shortfall is conservative.

Each rule reaches only what its number sees

“A rule built on platform records reaches the sellers on the platforms. In Indonesia that is fewer than one in five.”

Can be taxed: 5% of online sellers



On record, not taxed: 12%

Commission cap

Motorbike rides only (Gojek, Grab);
first test 27 October

Discount rule

Targets discounts the platforms were
already cutting

Marketplace tax

Can collect from about 1 seller in 20;
puts every marketplace seller on
record

2026 census

Reaches the same group the 2023
count found: the next jump may be
counting again

PMK 37/2025; Permendag 19/2026; Perpres 27/2026 as announced; BPS survey microdata. Abroad: OECD (2020), Directive 2021/514, BIR RR 16-2023, Decree 117/2025/ND-CP.

Answers to the questions from the oral

“Each question now has an answer in the thesis, not just at the podium.”

Prof. Thomas: what is the point of the wedge?

Its size sets how far a price change moves revenue growth. The wedge is why growth divergence is large in Indonesia.

Prof. Thomas: the ride-hailing commission cap

Scoped to motorbike rides; its slice of the measure is shown; predictions are scored on 27 October and in mid-November.

Prof. Thomas: politics and the players

The rules chapter maps each rule to the number it acts on and how far it reaches. No causal claims.

Prof. Kong: how transaction value compares across platforms

A definition map of each platform's transaction value, in the appendix.

Prof. Kong: hard to read, too cautious

A plain story with one line from start to end; the centre is a bolder, tested claim about the official count.

What we corrected along the way

“Each correction made a claim smaller and the thesis harder to knock over.”

BEFORE

“Indonesian platforms changed their prices far more”

“Mostly newly counted sellers”

“Markets did not pay up for fee-driven growth”

“Two thirds of revenue growth came from the cut”

“No one has compared platform and national figures”

NOW

Similar changes in rupiah; a much thinner slice

About half not explained by first-time entry (37-56%); returning sellers named

Market value moved with buying; the multiples comparison is arithmetic, not evidence

Two thirds of the movement, defined; not a share of net growth

UNCTAD (2024) has; our contribution is the connected Indonesian evidence

Next: writing, and four dates outside our control

"The material is complete. What remains is writing it and scoring the predictions."

27 Oct

1 Nov

Early Nov

Mid Nov



GoTo 3Q26: commission-cap predictions



Marketplace tax start (registered: slips again)



BPS 3Q26 GDP: buying vs household spending



Grab and Sea 3Q26: remaining predictions

WRITING

- Results chapter first, then the introduction
- Same chapters as the proposal; the story runs through them
- YZU template and reference check before submission

ONE QUESTION FOR YOU

H2 still holds on the published numbers, but its 2023 meaning changes: about half the rise in sellers was not new. Is it acceptable to present the entry test as closing the limit the proposal itself named?